

Out of all the concentrations big data is one that should be heavily emphasized due to its growth across all industries. I consistently see demand for data engineers in the workplace and giving students an education in this material would give them a huge edge over students that don't cover it.

At UOP, I found that coworkers who graduated from other universities had more rigorous coursework with process simulations (Aspen, Unisim, etc.) Would be useful to implement more simulation work for students wanting to pursue design engineering

I am very pleased to see the program expanding the selection of concentrations.

Machine Learning Algorithms will become basic standard tool for process modeling and process control.

My company has had a recent push toward water sustainability and it would be good to understand different water processing techniques in the industry. Understanding how food processing and chemical engineering interact could have been a more applicable learning for a lot of my peers who went into food processing.

electrochemistry as an elective part of the emerging markets

Students would probably appreciate not having PChem 2 as a requirement. I believe none of the other IL universities have that class as a requirement. I had to give up taking ASPEN because I couldn't handle both PChem 2 and ASPEN

My senior class was extremely divided socially. Classes should force group work that incorporates groups working with groups. My day to day work in my engineering group working with other engineering groups.

Speaking from my own experience in Polymers, I know that a concentration in Polymer Science or Material Science would be of much use to my current role as a Chemical Process Engineer.

Manufacturing Designing (Autocad/Solidworks)

Since entering the job market and searching for a job, I have not seen any demand whatsoever for a concentration.

Energy & Environment would definitely have been a concentration I would have pursued if it was offered while I was in school, and it would've greatly positively impacted my chances of finding a job in that industry. I know other students that would've also been interested in it.

In my opinions, students in our department lack experience in getting in touch with the outside world, they should go to more events like conferences and get in touch with the people from companies, or tours to companies will be helpful.

I think adding new concentrations is a great idea. My only concern would be if any of these courses would replace the core courses currently required.

There is a huge market for controls and automation engineers. Big data analytics is exploding as well. Marketing would help round out candidates from a business perspective.

Maybe a focus in battery technology? I am not sure if this is included in some other curriculum.

The importance I rated based on my interest. I did get a job in environmental technologies & they only hire ChEs. When i was lookin for jobs, Big Data jobs were very plentiful (esp in chicago) & I considered that avenue. Having this concentration I believe would be very marketable to chicago.

When I was job hunting, I've found out that a lot of chemical engineering positions are related to polymers, plastics, and polymer sciences. I think it would be good to give students more exposure to it.

Food technology and Environmental are huge. It's a bummer that most of the environmental engineering is done through civil or mechE and not ChemE. ChemE's, I feel, can make a bigger impact on this footprint going forward. Water treatment might be another one that can interest some ChemE students as well.

Production Engineering, Process Engineering(Process control, PI interface, HMI, process optimization, efficiency), Quality Control engineering(Out of spec product, linear and non linear blending), Compliance/EHS engineering, Product Engineering, Project management engineering(MOC- management of change, project charter, getting quotes, optimizing design, contractor safety, PJHA)

Automotive Technology - Vehicle and Power-train Systems - Battery (Li-Ion & Lead Acid)

Process Simulations and Energy & Environment concentrations seem to compliment well with the pharmaceutical and food industries, but more detail is needed behind it to ensure the concentration tackles common problems such as software limitations and legal concerns.

i think that these would be very helpful, particularly adding a pharmaceutical concentration because of the amount of pharmaceutical companies located in the chicagoland area. it would help UIC students to be more marketable to these companies and help them find jobs here at home.

More classes/electives that cover a wider range of careers such as classes related to engineering sales, programming, nuclear energy etc. Lots of ChemE students at UIC dont know or are not qualified enough to find these types of jobs so offering extra course in these areas can expose students to that. Classes like Organic Chem 2 can be removed as a requirement to make room in the schedule for undergrads.

To give context to my responses: my position is currently in the pharmaceutical industry.

Include contract law on entre/IP. Any engineer going into a design firm or EPC will eventually deal with. Project scheduling (software and theory) could be another benefit for engineers going into project management

I work in pharmaceuticals, but I see that arm being international. In my world, molecular simulation and big data are the largest factors. But employable graduates come from sustainability/energy/environment. Nanotechnology will be a strong future area of interest

I think these concentrations would be very beneficial.

Things are shifting to technology and e-commerce. Students need soft skills to market themselves. Leadership is becoming the top characteristic employers look for; someone who can work well with people on top of their technical skills.

Every 300 level course should have some sort of lab to go along with the material. I didn't know what an automated valve or a rupture disk looked like until i started working in a plant.

In utilities / every sector. Some bias might appear

Any cross-collaborative effort with other pure or applied science department will always yield good results!

My job values statistical analysis, so adding that as a requirement would be very beneficial.

Manufacturing Engineering would be helpful. Many of the fresh college graduates we hire at Kazmira have not had experience in field work beyond unit operations.

To be honest, my responses are lean favorably on nanotechnology because of past industrial gas experience in the semiconductor industry. In my line of work, I visited with a several university with nano fabs and envy their research capabilities, wishing to see that at UIC in the future.

The most of courses are also included in other majors, as a result students can be encouraged to take classes for big data, food technology, material science from other majors. However, process simulation can make difference for ChemE students.

I think the ChemE program should teach some general engineering basics such as how to read and interpret engineer drawings and industry specifications.